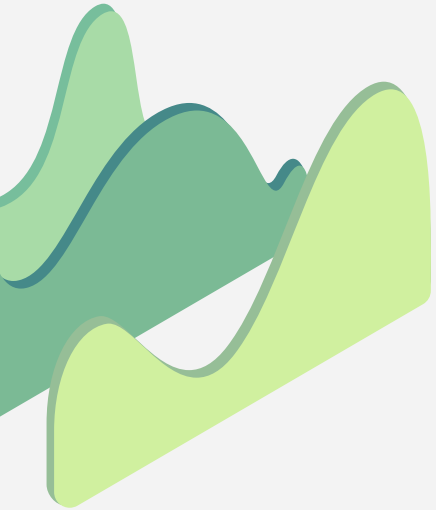
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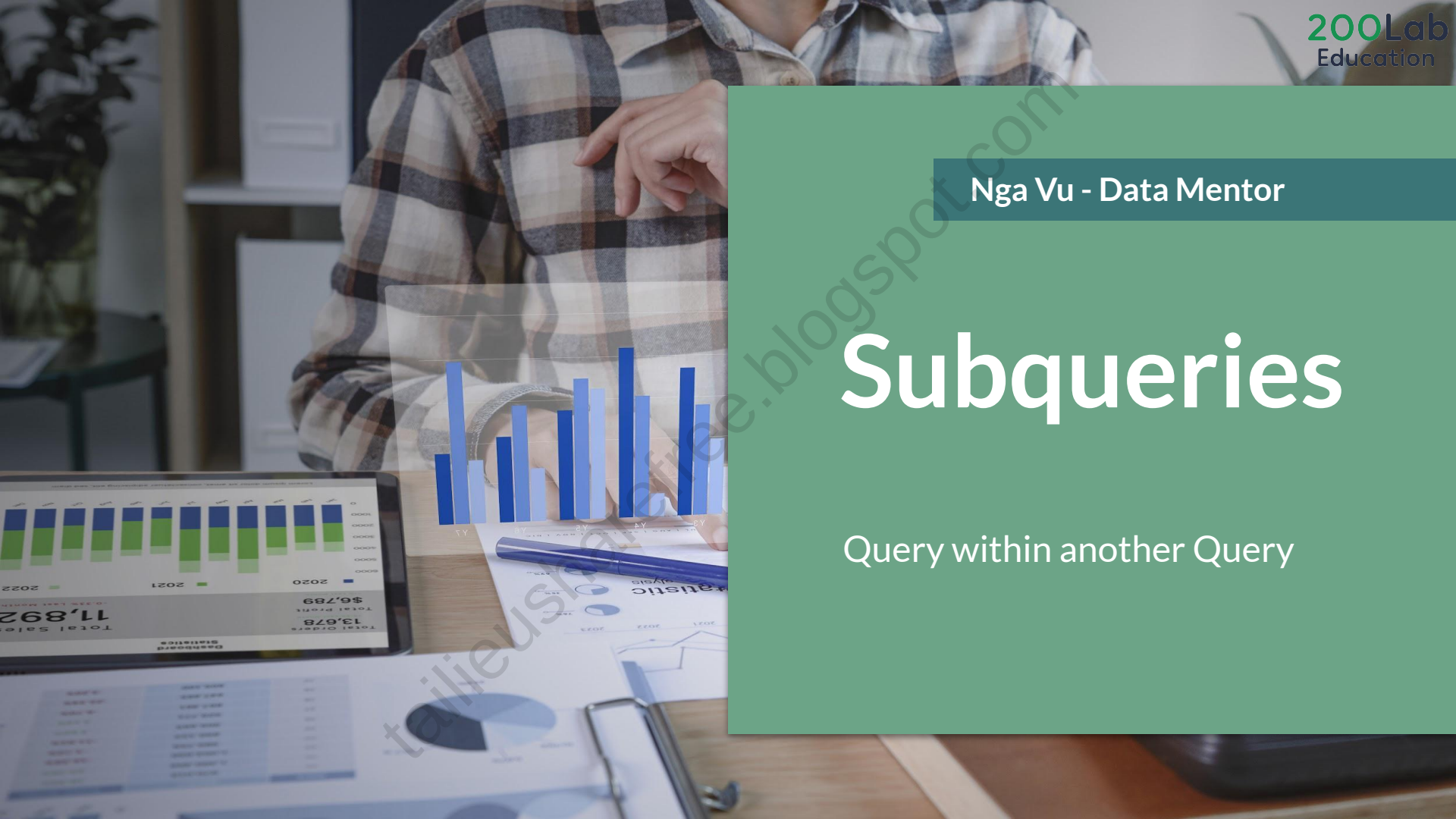
SQL for Beginners

Nga Vu - Data Mentor





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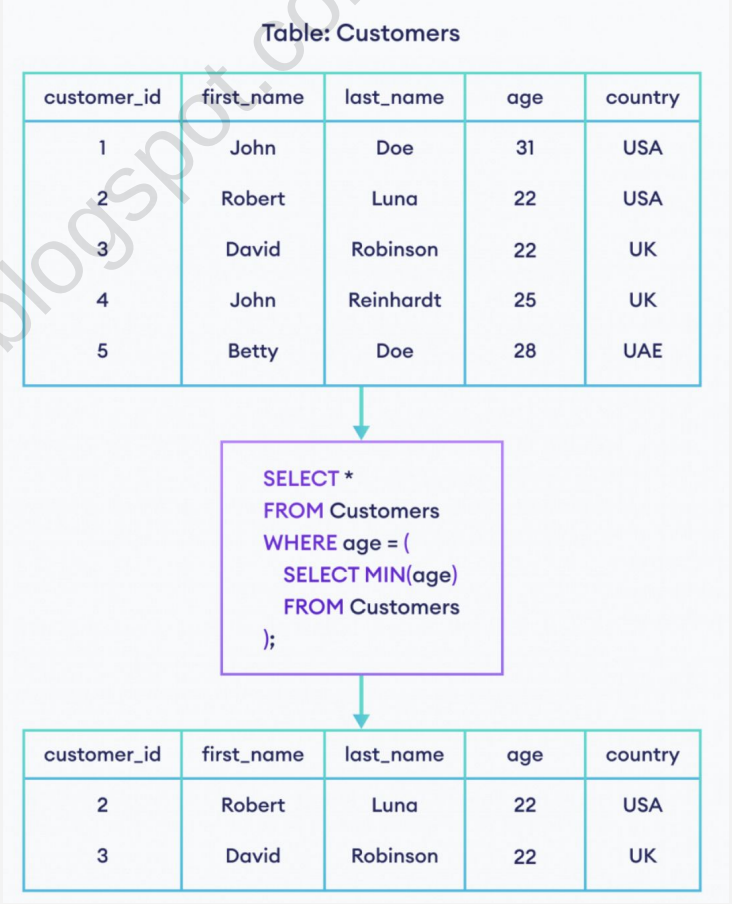
Subqueries

Query within another Query

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Subquery - Definition

- **Subquery** can be simply defined as a query within another query
- You can place the Subquery in: **WHERE** clause, **HAVING** clause, **FROM** clause
- The outer query is called as **main query** and inner query is called as **subquery**
- The subquery generally executes first when the subquery doesn't have any **correlation** with the main query



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Subquery - Syntax

FROM Clause

```
SELECT column_name
FROM ( SELECT COLUMN_NAME  from
TABLE_NAME   WHERE ... ) alias_name
```

```
SELECT *
FROM ( SELECT customer_id, email
       FROM customer
       WHERE store_id = 1 ) cus_sub
```

Commonly used with **JOIN**

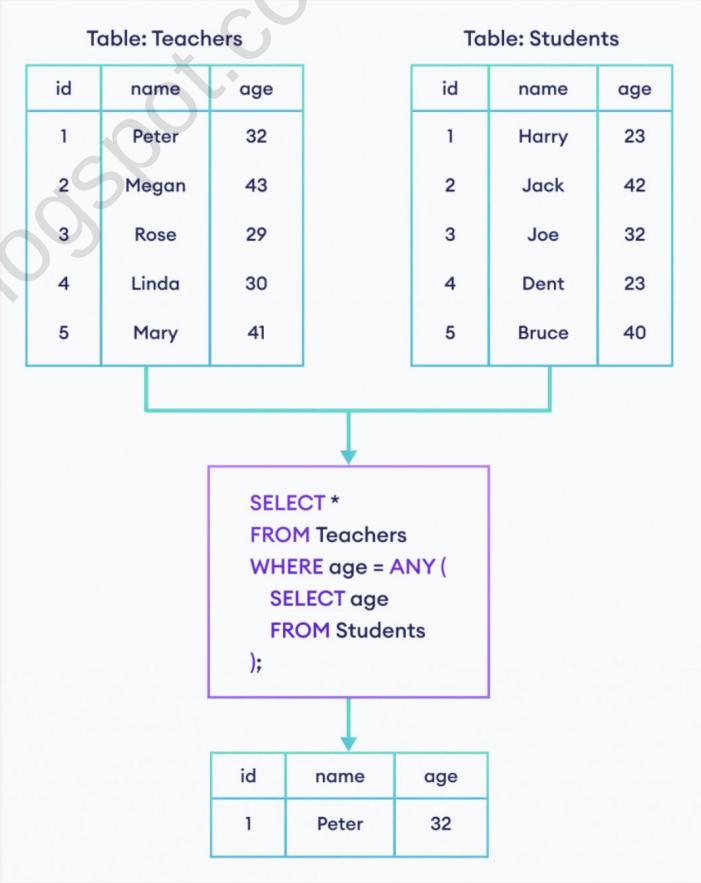
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Subquery - Syntax

WHERE Clause

SELECT column_name
FROM table_name
WHERE column_name
[comparison_operator **ANY** (subquery)
IN (subquery)
comparison_operator **SOME** (subquery)]

comparison_operator: = > < >= <= <> !=



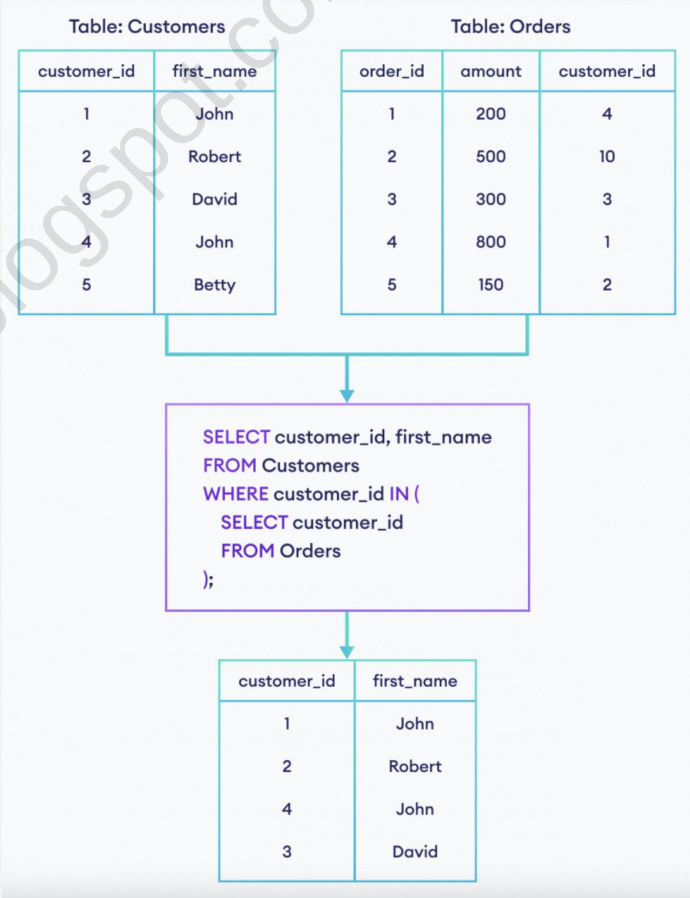
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Subquery - Syntax

WHERE Clause

```
SELECT column_name
FROM table_name
WHERE column_name
[comparison_operator ANY (subquery)
IN (subquery)
comparison_operator SOME (subquery)]
```

```
SELECT *
FROM customers c
WHERE customerNumber IN (
        SELECT DISTINCT customerNumber
        FROM orders)
```



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Subquery - Syntax

HAVING Clause

```
SELECT r.customer_id, COUNT(*) no_rental
FROM sakila.rental r
GROUP BY 1
HAVING no_rental >= (SELECT AVG(no_rental)
      FROM (SELECT r.customer_id, COUNT(*)
no_rental
        FROM sakila.rental r
        GROUP BY 1) cus_rental)
```

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Types of Subquery

There are 4 types:

1. **Scalar**: return a single value
2. Single Row: return a single row
3. **Column**: return a single/multiple column(s)
4. Correlated: depend on data provided by the outer query

```
SELECT *
FROM sakila.rental r
WHERE customer_id  IN
      (SELECT customer_id
      FROM sakila.customer c
      WHERE c.active = 1)
```

```
SELECT r.customer_id, COUNT(*) no_rental
FROM sakila.rental r
GROUP BY 1
HAVING no_rental >= (SELECT AVG(no_rental)
      FROM (SELECT r.customer_id, COUNT(*) no_rental
       FROM sakila.rental r
       GROUP BY 1) cus_rental)
```

```
SELECT *

FROM ecommerce.orderdetails o

WHERE (orderNumber,productCode) IN (

      SELECT orderNumber ,productCode

      FROM ecommerce.orderdetails o2

      WHERE o2.orderNumber =10100)
```

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Types of Subquery

There are 4 types:

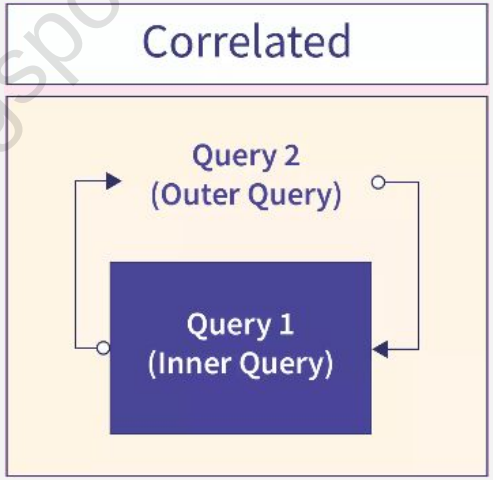
1. Scalar: return a single value
2. **Single Row**: return a single row
3. Column: return a single/multiple column(s)
4. Correlated: depend on data provided by the outer query

```
SELECT *
FROM ecommerce.orderdetails o
WHERE ROW(o.orderNumber,o.productCode) =
      (SELECT '10100','S18_1749')
```

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Correlated Subquery

- Correlated subquery is one where the inner query depends on values provided by the outer query
- This means that in a correlated subquery,the inner query is executed repeatedly, once for each row that might be selected by the outer query



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Correlated Subquery

Example

```
SELECT last_name, salary, department_id
FROM employees outer
WHERE salary >
        (SELECT AVG(salary)
            FROM employees
            WHERE department_id =
            outer.department_id);
```

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Correlated Subquery

Example

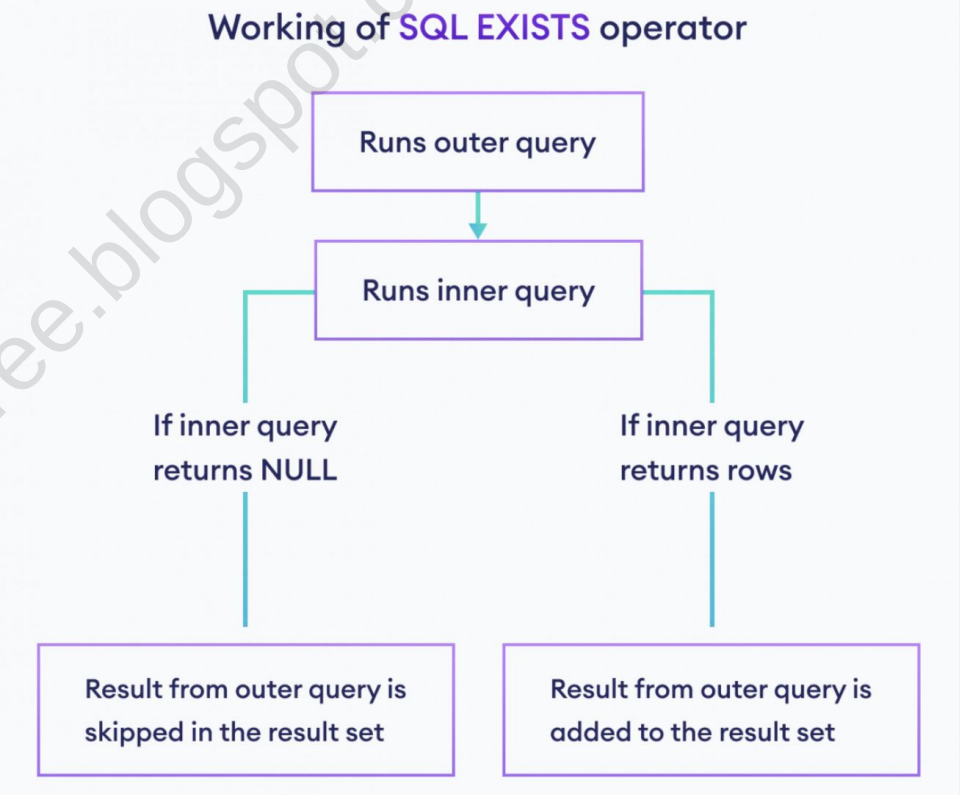
```
SELECT c.name , f.title , f.replacement_cost
FROM film f
        JOIN film_category fc ON fc.film_id = f.film_id
        JOIN category c ON c.category_id = fc.category_id
WHERE f.replacement_cost =
(SELECT MIN(replacement_cost)
                FROM film f2 JOIN film_category fc2
                ON fc2.film_id = f2.film_id
                WHERE fc2.category_id = fc.category_id )
ORDER BY 1
```

	name	title	replacement_cost
1	Action	KISSING DOLLS	9.99
2	Action	TRUMAN CRAZY	9.99
3	Animation	ANACONDA CONFESSIONS	9.99
4	Animation	TELEMARK HEARTBREAKERS	9.99
5	Animation	THEORY MERMAID	9.99
6	Animation	WAIT CIDER	9.99
7	Children	HEARTBREAKERS BRIGHT	9.99
8	Classics	CANDIDATE PERDITION	10.99
9	Classics	OCTOBER SUBMARINE	10.99
10	Classics	SLING LUKE	10.99

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Correlated Subquery : EXISTS vs NOT EXISTS

- The EXISTS operator is used to test for the existence of any record in a **subquery**.
- The EXISTS operator returns TRUE if the subquery returns one or more records.
- NOT EXISTS returns TRUE if the result of the subquery does not contain any rows



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Correlated Subquery : EXISTS vs NOT EXISTS

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Example

```
SELECT emp_no ,d.dept_name

FROM employees.dept_emp de JOIN
employees.departments d ON d.dept_no =
de.dept_no

WHERE d.dept_name IN
('Production','Quality Management')

ORDER BY 1
```

	emp_no	dept_name
1	10,003	Production
2	10,004	Production
3	10,009	Quality Management
4	10,010	Production
5	10,010	Quality Management
6	10,018	Production
7	10,020	Production
8	10,024	Production
9	10,026	Production
10	10,029	Production



```
SELECT emp_no ,d.dept_name

FROM employees.dept_emp de

        JOIN employees.departments d ON
d.dept_no = de.dept_no

WHERE d.dept_name ='Production'

        AND d.dept_name ='Quality Management'


ORDER BY 1
```

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Correlated Subquery : EXISTS vs NOT EXISTS

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Example

```
SELECT emp_no ,d.dept_name

FROM employees.dept_emp de

        JOIN employees.departments d ON d.dept_no = de.dept_no

WHERE d.dept_name ='Production'

        AND EXISTS ( SELECT 1

                FROM employees.dept_emp de1

                        JOIN employees.departments d1 ON d1.dept_no =
de1.dept_no

                WHERE de1.emp_no=de.emp_no

                    AND d1.dept_name='Quality Management')

ORDER BY 1
```

```
SELECT COUNT(DISTINCT de.emp_no)

FROM employees.dept_emp de

        JOIN employees.departments d ON d.dept_no
= de.dept_no

WHERE d.dept_name ='Production'

        AND EXISTS ( SELECT 1

                FROM employees.dept_emp de1

                        JOIN employees.departments
d1 ON d1.dept_no = de1.dept_no

                WHERE de1.emp_no=de.emp_no

                    AND d1.dept_name='Quality
Management')
```

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When Subquery ?

- Almost subqueries can be converted to join
- Use subqueries when the result that you want requires more than one query and each subquery provides a subset of the table involved in the query
- Avoid using correlated subquery on large result set, use join instead



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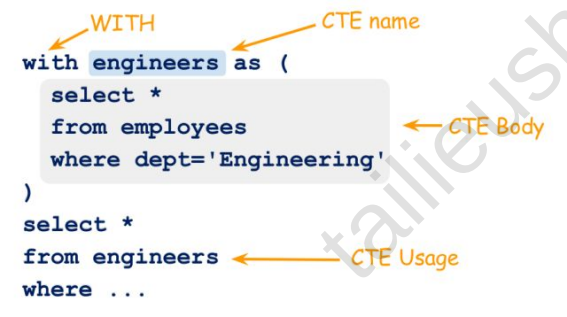
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Common Table Expression

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CTE - Definition

- A common table expression (CTE) is a named temporary result set that exists within the scope of a single statement.
- It can be referred to later within that statement, possibly multiple time




```
SELECT r.customer_id, COUNT(*) no_rental
FROM sakila.rental r
GROUP BY 1
HAVING no_rental >= (SELECT AVG(no_rental)
      FROM (SELECT r.customer_id, COUNT(*) no_rental
       FROM sakila.rental r
       GROUP BY 1) cus_rental)
```

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Why CTE ?

- Easily write and maintain complex queries via increased readability and simplification
- Reference the resulting table multiple times in the same statement
- Save result of query for later use

```
WITH cte_rental_sum AS
(SELECT r.customer_id, COUNT(*) no_rental
FROM sakila.rental r
GROUP BY 1)

SELECT *
FROM cte_rental_sum
WHERE no_rental >= (SELECT AVG(no_rental)
                    FROM cte_rental_sum )
```

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